Assignment 1: ELL7286 (Introduction to Machine Learning)

Jyotirmoy Nath- 2025EEZ8127

Suresh bushetty- 2025EEZ8171

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Part 1: Video Summarization

1. On What Grounds is a Keyframe Segregated from a Non-Informative Frame?

A keyframe is identified as a frame that best represents the dominant visual and temporal dynamics of a video. Intuitively, informative frames correspond to moments where meaningful changes occur, while non-informative frames correspond to redundant or static content.

Fundamental Principle
Let consecutive frames be denoted by F_t and F_{t-1}. Motion is defined as:

$$M_t = |F_t - F_{t-1}|$$

Frames with larger motion magnitude contain stronger visual change and are more likely to represent important events.

Grounds for Segregation

1. **Motion Significance:** Frames showing strong temporal variation indicate activity, transitions, or new visual information. Frames with negligible change correspond to static or repetitive content and are treated as non-informative.

2. **Variance Contribution via PCA:** Principal Component Analysis identifies dominant directions of variation in the video. Frames that project strongly onto leading components capture major structural or motion patterns and therefore contain higher information content.

3. **Quantitative Importance Score:** Each frame is assigned a score based on its weighted projection onto principal components. Higher scores indicate stronger contribution to global variance and thus higher representational importance.

4. **Redundancy Suppression:** Even if a frame has a high score, it is rejected if it is visually similar to neighbouring frames. This prevents multiple selections representing the same event.

5. **Temporal Representativeness:** Informative frames must collectively span the entire timeline so that all major events are represented. Frames clustered within a short interval are penalised.

Decision Rule A frame is selected as a keyframe if it simultaneously satisfies:

- high motion magnitude,
- strong PCA contribution,
- low redundancy with adjacent frames, and
- improvement in temporal coverage.

Frames failing these conditions are classified as non-informative because they do not significantly contribute new information to the video summary.

2. Model Used for Video Summarisation

The proposed model is eigen-analysis based video summarisation framework that selects informative frames using motion-driven Principal Component Analysis (PCA).

2.1 Pipeline

The summarisation process consists of the following stages:

1. Extract frames from the video
2. Compute motion representations using frame differencing
3. Construct a feature matrix and apply PCA for dimensionality reduction
4. Compute importance scores for each frame
5. Select keyframes using score ranking with temporal coverage constraints
6. Generate the final summarised video from selected frames

This model treats video summarisation as a variance maximisation problem: frames that best explain the dominant variability of the sequence are retained, while redundant frames are discarded.

3. Experimentation with Different Numbers of Frames

To analyze the effect of summary length, experiments were conducted with different numbers of selected keyframes (N). The results are shown below:

N	Score	Coverage (%)	Compression
5	0.6319	19.0	199.8×
10	0.3679	25.0	99.9×
15	0.2583	67.5	66.6×
20	0.2009	77.0	50.0×
25	0.1652	77.0	40.0×
30	0.1409	77.0	33.3×
40	0.1093	96.0	25.0×
50	0.0894	99.5	20.0×

Observations

- Increasing N improves temporal coverage of the video.

- Average frame score decreases as more frames are added, since lower-importance frames begin to enter the summary.
- Compression decreases with larger N, indicating a trade-off between compactness and completeness.
- Coverage saturates around $N = 20$–30, after which additional frames mainly increase redundancy rather than information.

4. Selection of Optimal Number of Keyframes

The optimal number of keyframes is determined by balancing three competing objectives: information preservation, temporal coverage, and compression efficiency. Increasing the number of frames improves coverage of events but reduces compression, while too few frames increase compression at the cost of missing important content. To select the optimal value, multiple candidate values of N are tested and evaluated using quantitative metrics. The optimal N is chosen where:

- Temporal coverage becomes sufficiently high
- Average frame importance score remains reasonably large
- Compression is still significant

5. Evaluation Measures for Summarisation Quality

The quality of the generated video summary is evaluated using quantitative metrics computed for the selected number of keyframes ($N = 20$). The results are:

Metric	Value
Average Score	0.2009
Coverage	77.0%
Compression Ratio	50.0×
Precision	90.0%
Recall	7.2%
F1 Score	13.33

6. Interpretation of the Output

- **Average Score:** Indicates that selected frames contain strong contributions to dominant motion patterns.
- **Coverage (77%):** Shows that the summary represents most temporal segments of the video.
- **Compression (50×):** Demonstrates significant reduction in data while retaining informative content.
- **Precision (90%):** Most selected frames are truly informative, meaning the selection strategy is accurate.
- **Recall (7.2%):** Only a small portion of all informative frames are selected, which is expected because summarisation intentionally keeps only a compact subset.

- **F1 Score:** Reflects the trade-off between precision and recall. A lower F1 is acceptable in summarisation tasks where compactness is prioritized over exhaustive inclusion.

Part 2: Background Subtraction

1. Method Used

Background subtraction is performed using motion segmentation followed by foreground extraction. Two algorithms were implemented from scratch for this task: Sequential Labeling and a pixel-wise Gaussian Mixture Model (GMM) for experimental comparison.

Motion is computed using frame differencing:

$$D_t = |F_t - F_{t-1}|$$

Pixels whose difference exceeds a threshold are classified as moving.

Sequential Labeling (Primary Method) The binary motion masks are processed using connected component analysis to group neighboring moving pixels into regions. Small regions are removed as noise using an area threshold. The remaining components represent foreground objects.

Parameters used

- Threshold = 2.2
- Minimum region area = 50 pixels

Gaussian Mixture Model (Experimental Method) A Gaussian Mixture Model is implemented to model each pixel as a mixture of K Gaussian distributions. For each frame:

- Pixel values are matched against existing Gaussians
- Parameters are updated using learning rate α
- Components are ranked by weight/variance
- Top components represent background
- Remaining pixels are labeled foreground

Base Configuration

- $K = 3$
- $\alpha = 0.01$
- Threshold = 2.2
- Background ratio = 0.7
- Initial variance = 15

Experimental Comparison of GMM Parameters Different configurations were tested to analyze sensitivity:

Configuration	Foreground Percentage
$K = 3,\ \alpha = 0.01,\ T = 2.5$	8.81%
$K = 3,\ \alpha = 0.02,\ T = 2.5$	6.63%
$K = 4,\ \alpha = 0.01,\ T = 2.5$	8.03%
$K = 3,\ \alpha = 0.01,\ T = 3.0$	8.03%

2. Generated Outputs

The system produces:

- Background video reconstructed from background pixels
- Foreground video containing only moving objects
- ASCII parameter file documenting all algorithm settings

Part 3: Background Subtraction in Summarised Video

1. Parameter Tuning

Summarised videos have larger temporal gaps between frames, which increases apparent motion magnitude. Therefore, stricter parameters are required to prevent over-segmentation.

Final Parameters Used

Parameter	(i) Full Video	(ii) Summarised Video
Threshold	2.2	10.0
Min Area	50	100

2. Comparison: Full vs Summarised Video

Metric	(i) Full Video	(ii) Summary Video
Frames	998	19
Avg FG Ratio	0.0779	0.0570
Consistency	0.9317	0.9696

Observations

- Summarisation reduces frames by $\sim 98\%$, greatly improving processing speed.
- Average foreground ratio decreases, indicating removal of redundant motion.
- Consistency increases, meaning detected motion regions are more stable across frames.